Integral comparison test — 3.7

Recall that if $0 \le f(x) \le g(x)$, then

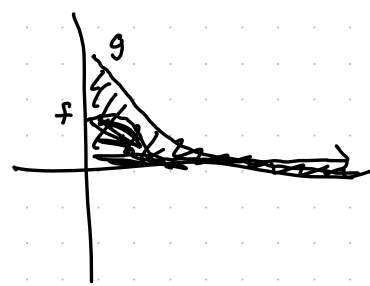
$$\int_a^b f(x)\,dx \le \int_a^b g(x)\,dx$$

and this continues to hold if the integrals are improper.

<u>Comparison test:</u> If $0 \le f(x) \le g(x)$, then

$$\int_a^b f(x)\,dx \text{ diverges} \implies \int_a^b g(x)\,dx \text{ diverges}$$

$$\int_a^b g(x)\,dx \text{ converges} \implies \int_a^b f(x)\,dx \text{ converges}$$



Example: Does $\int_1^\infty \frac{1}{xe^x}\,dx$ converge or diverge?

$\frac{1}{xe^x} \le \frac{1}{e^x} \implies \int_1^\infty \frac{1}{xe^x}\,dx$ converges if we can show that $\int_1^\infty \frac{1}{e^x}\,dx$ converges.

$$\int_1^\infty \frac{1}{e^x}\,dx = \lim_{b\to\infty} \int_1^b e^{-x}\,dx = \lim_{b\to\infty} -e^{-x}\Big|_1^b$$

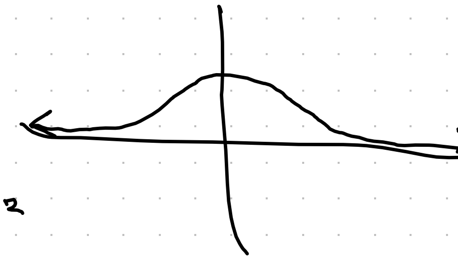
$$= \lim_{b\to\infty} \frac{-1}{e^b} + \frac{1}{e} = \frac{1}{e}$$

Don't know exactly what $\int_1^\infty \frac{1}{xe^x}\,dx$ is, but it is $\le \frac{1}{e}$ by the comparison test.

(It converges.)

<u>Example</u>: $\int_{-\infty}^{\infty} e^{-x^2}\,dx$ converges ($\text{it is } \sqrt{\pi}$)

$$= \int_{-\infty}^{0} e^{-x^2}\,dx + \int_{0}^{\infty} e^{-x^2}\,dx$$



Note that on $[1,\infty)$, $e^{-x^2} \leq x e^{-x^2}$

and $\int_1^{\infty} x e^{-x^2}\,dx = \lim\limits_{b\to\infty} \int_1^b x e^{-x^2}\,dx$ $\quad u = x^2$, $du = 2x\,dx$

$$= \frac{1}{2} \lim_{b\to\infty} \int_1^{b^2} e^{-u}\,du = \frac{1}{2}\left(\lim_{b\to\infty} -e^{-b^2} + e^{-1}\right) = \frac{1}{2e} \quad \text{finite}$$

$\int_1^{\infty} x e^{-x^2}\,dx$ converges $\Rightarrow \int_1^{\infty} e^{-x^2}\,dx$ converges.

$\Rightarrow \int_0^{\infty} e^{-x^2}\,dx$ converges. By a similar argument,

or by symmetry, $\int_{-\infty}^{0} e^{-x^2}\,dx$ converges as well